

LABORATORY MANUAL - THE SCIENCE OF SOUND



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Laboratory Manual - The Science of Sound

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Licensing

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CHAPTER OVERVIEW

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4.1.1: Introduction

In this exercise, force and pressure are examined through direct observation in a qualitative manner. Force may be computed by multiplying the mass of an object by acceleration:

$$f = ma$$

If the mass is measured in kilograms and the acceleration in meters per second squared, the resulting unit of force is kilogram meters per second squared, or *newtons* (N). For comparison, one pound of force is equivalent to approximately 4.45 N. A mass sitting on the Earth is affected by Earth's gravitational acceleration, 9.8 m/s^2 , so the force acting on it at rest (i.e., its weight) is the mass in kg times 9.8 m/s^2 .

Pressure is defined as force per unit area:

$$p = f/\text{area}$$

Given a force in Newtons and an area in square meters, the resulting unit (Newtons per square meter) is called a *Pascal* (Pa). Note that pressure can be increased by either increasing the force or by decreasing the area. Even a modest force can produce extremely high pressure if it is applied over an exceedingly small area.

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4.1.2: Equipment

- 1 Meter stick or tape measure
 - 1 Metric mass set
 - 1 Metal plate
 - 1 Sheet of foam rubber
-

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4.1.3: Procedure

Part A: Calculating Force and Pressure

1. Calculate the expected force for the 1000 g (1 kilogram) mass based on the acceleration due to Earth's gravity and record in Table 1.
2. Repeat step one for the 500 g and 200 g masses.
3. Measure the diameter of the 1000 g mass. Calculate the resulting area ($area = \pi r^2$) and record in table two. Using the force computed in Table 1, calculate and enter the resulting pressure in Table 2.
4. Repeat step three for the 500 g and 200 g masses.

Part B: Observing Pressure via Deformation

5. A simple qualitative way of observing pressure is to note the deformation of a semi-soft material. In this exercise, a slab of foam rubber will serve this purpose. Simply place the 1000 g mass on the slab and note the degree of deformation (none, slight, moderate, etc.). Enter your observation in the final column of Table 2
6. Repeat step five for the 500 g and 200 g masses.
7. Using the same force, pressure may be increased or decreased by decreasing or increasing the area, respectively. To increase the effective area, a metal plate may be introduced between the mass and foam rubber. Calculate the area of the plate, and based on this value, calculate the new pressures for the three masses and enter these data in Table 3.
8. Place the metal plate on the foam rubber and position the 1000 g mass in its center. Note the degree of deformation of the foam rubber and record in Table 3.
9. Repeat step five for the 500 g and 200 g masses.
10. To illustrate the effect of decreasing the area, place the plate on the foam rubber on edge, that is, the plate should be perpendicular to the foam rubber. Carefully balance the 1000 g mass on the top edge of the plate (you will have to steady it with your hands) and note the deformation. The area in this instance is the area of the edge of the plate, and thus very, very small compared to the plate face or the bottom of the mass. (This observation is sufficient and there is no need to attempt to measure the area of this edge.)

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4.1.4: Data Tables

Table 1

Mass	Force
1000 g	
500 g	
200 g	

Table 2

Mass	Diameter	Area	Pressure	Observations
1000 g				
500 g				
200 g				

Table 3

Mass	Plate Area	Pressure	Observations
1000 g			
500 g			
200 g			

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4.1.5: Questions

1. What is the affect of increasing mass on force?
 2. What is the affect of increasing area on pressure?
 3. Does the higher of two masses always produce a higher force given the same acceleration?
 4. Does the higher of two masses always produce a higher pressure?
 5. Based on established theory, if this exercise had been repeated on the moon, would you expect the deformations recorded in Tables 2 and 3 to be the same, greater, or smaller? Why?
-

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CHAPTER OVERVIEW

4.2: Speed of Sound

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4.2.1: Introduction

The speed of sound in air at room temperature is approximately 343 meters per second, or 1125 feet per second, and increases with increasing temperature. This velocity is roughly equivalent to approximately 1 foot every 0.9 milliseconds. Thus, for the dimensions of typical class and lab rooms, time delays are generally less than 50 milliseconds, and are therefore difficult to measure without specialized equipment. Digital oscilloscopes can easily resolve time differences several orders of magnitude less than one millisecond, and thus are a good candidate for this exercise. Care must be taken to make sure that no differential delay is caused by the recording circuitry.

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